EIMAJD Automation

System specification book

Industrial poultry-house monitoring, shared production records and supervised pilot commands.

REVISION	ISSUED	IMPLEMENTATION BASELINE
0.6	07 November 2026	Android 1.13.1 (29) Opta 4.1.0-feed-chains

Controlled working baseline. This revision reconciles approved requirements with development evidence. Software delivery is not field commissioning, business acceptance or production release approval.

Pilot: one farm, one poultry house, one Arduino Opta, one AFX00005 expansion and two users. Owner/admin: Mahdi Guermazi. Manager: Adnen Guermazi.

Supersedes specification v0.5 of 06 November 2026 for requirements and status.

01 / GOVERNANCE

Document control and revision

Purpose: provide one current functional and technical baseline for the developed system, including implemented behavior, approved rules, known limitations and acceptance gates.

Item	Controlled value
Document owner	Mahdi Guermazi, owner/admin
Operational pilot manager	Adnen Guermazi
Review date / cut-off	07 November 2026. Implementation evidence through 8445db3.
Working branch	Opta_Feed_Chain_Expansion
Application build evidence	Source 8445db3. GitHub Actions 42356825403.
Issue status	Working baseline updated from development; no new owner acceptance is inferred
Secret handling	No passwords, access/refresh tokens, private keys or real production totals are included

Revision history

Version	Change
0.1	Initial system requirements and conceptual blueprint.
0.2	Supervised command scope, pilot I/O, business rules and roles clarified.
0.3	Manager editing, fixed GMT+1 and carton receipts clarified; alarm policy proposed.
0.4	Records the implemented cloud path, alarms, FCM/Gmail, six-tab Android UI, business ledgers, password recovery and remaining acceptance gates.
0.5	Adds AFX00005 feed chain monitoring and field acceptance. Updates five-minute history and command recovery evidence.
0.6	Adds live 3D visualisation, corrected fan layout and camera, blue cooling pads, exact signal mapping, freshness and device acceptance. Updates Android to 1.13.1.

Status terms: APPROVED = agreed requirement. IMPLEMENTED = source/deployment evidence exists. VERIFIED = stated test evidence exists. PENDING = acceptance or implementation is still missing. These terms are not interchangeable.

02 / NAVIGATION

Reading guide and contents

Use the requirements sections to assess behavior; use the evidence and acceptance sections to decide whether a milestone can close. The PDF is the review copy; the companion Markdown and JSON preserve the editable baseline.

Pages	Subject
4	Scope, operating limits and terminology
5-6	Architecture and communication interfaces
7-8	Onboard I/O and AFX00005 feed chain feedback
9-11	Firmware, telemetry contract and time/resilience
12-13	Cloud data and Android screens
14-16	Live 3D requirements, fan mapping and reference views
17	Accounts, sessions and permissions
18-20	Production, flock changes and carton inventory
21-22	Alarm matrix and notification workflow
23-24	Security, release, reliability and operations
25-26	Evidence matrix and phase roadmap
27	Acceptance requirements
28	Open issues and change control
29	References and delivery identifiers

Precedence

The owner's later confirmed decisions take precedence over earlier proposals. This revision uses the latest delivery report and actual implementation sources when older phase reports describe historical pending work. A difference between an approved requirement and current code is recorded as a gap, not silently treated as approval.

Traceability

Requirement identifiers in this book group related behaviors. Repository paths and test names on pages 25 and 29 provide implementation evidence. ClickUp remains the work-tracking system; its task statuses do not replace signed or recorded acceptance.

> All development and acceptance dates in this issue are explicit. Production-day calculations use fixed GMT+1 year-round; audit/telemetry timestamps use UTC.

03 / REQUIREMENTS BASELINE

Purpose, scope and definitions

- SCP-01: Monitor temperature, three water cisterns, two feed/alveole silos, cooling auxiliary feedback eight displayed fan/control indicators and 12 feed chain running feedback signals.

- SCP-02: Store validated telemetry and shared production, mortality, flock, packaging, alarm and audit records in Supabase.

- SCP-03: Provide Android access now; retain the secured web dashboard as Phase 6 work, not completed Android/web parity.

- SCP-04: Continue server alarm evaluation independently of whether an Android app is open.

- SCP-05: Retain supervised test commands with plain MQTT strings, QoS 0 and retain off.

Explicit exclusions

Automatic environmental control, unattended operational switching, independent fan-failure detection, SMS delivery, power-loss-safe device buffering and production-approved remote control are not delivered by this pilot.

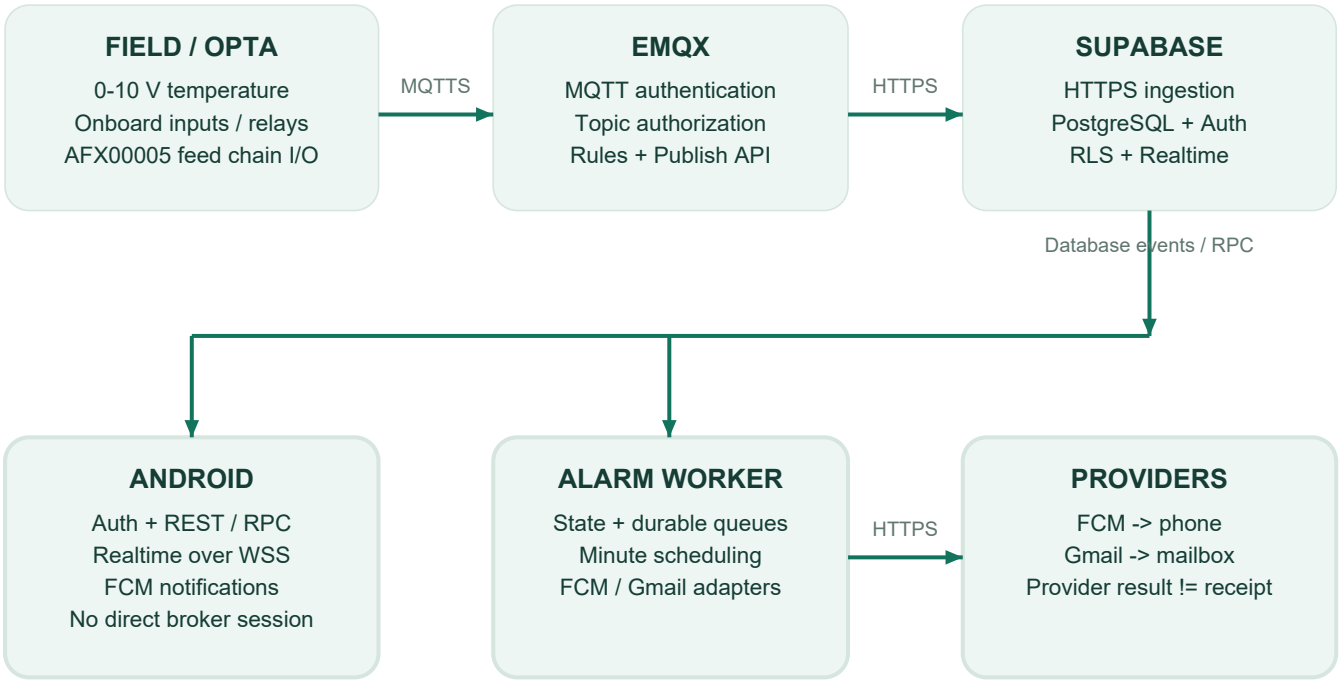
Term	Meaning
Stock / Sortie	Eggs in storage / eggs dispatched during the production day.
Production / rate	Calculated eggs / calculated eggs divided by the day's remaining flock.
Full / empty	Binary sensor interpretation. The fill illustration is not a measured percentage or volume.
Fan/control state	Relay or LED command/indication state, not confirmed fan rotation.
Cooling working	I7 auxiliary contact active; no water-flow or pressure measurement is claimed.
Feed chain running	High signal on the assigned AFX00005 digital input confirms chain running feedback.
Sent / delivered	Provider or broker acceptance differs from physical execution or human receipt.

Pumps, lighting, additional houses and independent fan rotation feedback remain future work. Existing schema support for membership across farms is not evidence of full multi-device command routing.

04 / SYSTEM DESIGN

Implemented system architecture

ARC-01: The Opta talks MQTT to EMQX. EMQX forwards telemetry through an HTTPS webhook. Supabase is the shared source of truth; Android uses Supabase Auth, REST/RPC and Realtime.



Command return path: Android -> Supabase -> EMQX HTTPS API -> Opta MQTT.

Four separate responsibilities

- Opta: sample onboard and AFX00005 inputs, convert temperature, timestamp/buffer frames and apply received pilot commands.

- EMQX: authenticate MQTT clients, authorize topics, route messages, forward telemetry and accept server-side publish requests.

- Supabase: validate identity/schema, deduplicate history, maintain latest state and five-minute chart readings, enforce business rules, evaluate alarms and audit delivery.

- Android: display authorized cloud data and render the live 3D model from the existing native telemetry events, submit validated business/configuration actions and show native push notifications.

Supabase does not run as a permanent MQTT subscriber in this deployment. Short-lived Edge Functions receive HTTP requests; no extra always-on MQTT bridge service is required. The public sender-information website is not the secured farm dashboard.

05 / INTERFACE CONTRACT

Communication interfaces

Interface	Transport / format	Authentication / purpose
Sensors -> Opta	0-10 V analog; 24 V binary electrical signals	Physical installation; no network authentication.
AFX00005 -> Opta	Opta AUX expansion bus	First expansion slot. Digital running feedback for 12 chains.
Opta <-> EMQX	MQTT over TLS, port 8883; Wi-Fi in current firmware	Device MQTT username/password; unique registered client ID.
Telemetry / status	opta/outputs JSON; opta/status online/offline text	Telemetry QoS 0, retain false. Status/LWT is retained; the command retain rule does not apply to status.
EMQX -> ingest-mqtt	HTTPS POST; topic, clientid and payload envelope	Protected x-emqx-webhook-secret header; enabled device/topic registry.
Android <-> Supabase	HTTPS Auth, REST/RPC; WSS Realtime	Supabase user access token, farm membership and RLS.
Android -> command function	HTTPS JSON request containing a payload string	Validated user session and role; command audit.
Command function -> EMQX	HTTPS /api/v5/publish	Server-only EMQX API identity/secret.
EMQX -> Opta commands	opta/cmd; plain text such as D0 ON	MQTT subscription; QoS 0 and retain false, fixed server-side.
Alarm worker -> providers	HTTPS FCM HTTP v1 / Gmail API	Protected service-account / mailbox OAuth credentials.
Opta -> NTP	DNS, then UDP/123; local UDP port 2390	Public time service; validates response before accepting UTC.

Pilot identifiers

Farm slug elmajd-poultry; house slug house-01; device and MQTT client ID opta-control-01. Supabase project zlmaombpfdznambrgqiv; Firebase project elmajd-automation. These identifiers are not passwords.

The proposed elmajd/{farm}/{house}/{device}/... topic hierarchy is future multi-device routing work. The existing pilot still uses opta/outputs, opta/status and opta/cmd.

06 / FIELD REQUIREMENTS

Hardware and logical I/O schedule

HW-01. The onboard Opta inputs keep the assignments below. The AFX00005 provides a separate set of terminals for feed chain running feedback, mapped on the next page.

Channel	Signal / equipment	Meaning / telemetry
I1 / A0	PT100 via 0-10 V transmitter	i1.volts, i1.temp_c, i1.fault; analog, not digital.
I2 / A1	24 V PNP/NO; cistern 1	inputs.I2; 1 full/detected, 0 empty.
I3 / A2	24 V PNP/NO; cistern 2	inputs.I3; same binary meaning.
I4 / A3	24 V PNP/NO; cistern 3	inputs.I4; same binary meaning.
I5 / A4	24 V PNP/NO; silo 1	inputs.I5; 1 material detected, 0 empty.
I6 / A5	24 V PNP/NO; silo 2	inputs.I6; same binary meaning.
I7 / A6	24 V NO auxiliary contact; cooling	inputs.I7; 1 working, 0 not working.
I8 / A7	Reserved	inputs.I8 transmitted; no approved equipment assignment.
D0-D3 / L0-L3	Four relay states + four LED states	Eight displayed control indicators; not eight independently confirmed fan-running inputs.

Temperature baseline and calibration

HW-02: Approved conversion is T(C) = -50 + 15 x input volts. Examples: 0 V = -50 C; 5 V = 25 C; 10 V = 100 C. The firmware uses 12-bit ADC sampling, 16-sample averaging and its approved divider/reference constants.

Required installed-chain accuracy remains +/-0.5 C, pending a trusted field comparison. The owner waived manufacturer/model paperwork as a pilot blocker; that waiver does not prove accuracy or replace electrical approval.

Final terminal numbers, cable/common/shield details, supply/protection/UPS sizing, grounding and installation-ready schematic remain pending. Level sensitivity is adjusted manually at each sensor.

07 / HARDWARE AND MONITORING

Feed chain feedback

FCH-01. Use one Arduino Opta AFX00005 (D1608E) digital expansion in the first slot immediately to the right of the base Opta. Monitor one chain per level across four rows. The terminals below are on the expansion.

Rangée	Étage 1	Étage 2	Étage 3
1	I1	I2	I3
2	I4	I5	I6
3	I7	I8	I9
4	I10	I11	I12

FCH-02. Each input receives a running feedback signal. High means En marche and low means À l'arrêt. I13-I16 are unassigned. Verify the signal interface and terminal mapping against the Arduino specifications and the final cabinet drawing.

Firmware and cloud contract

FCH-03. Read the expansion with Arduino_Opta_Blueprint 0.2.9 in the normal five-second capture cycle. Capture all input bits from one successful bus read. A missing module or failed read produces an unavailable reading. Capture and replay use the existing device identity and frame ordering.

FCH-04. Schema 2 adds optional feed_chains, a 12-bit integer from 0 to 4095. Bit 0 maps to I1 and bit 11 to I12. Zero represents 12 low inputs. JSON null means unavailable. Ingestion accepts earlier firmware that omits the field and converts malformed values to null. Supabase stores the reading and its receipt time in raw telemetry and source-ordered device_latest.

Android display and commissioning

FCH-05. In État > Chaînes d'aliment, display all 12 signals in four rows with three levels, input labels and text states. Show Indisponible for missing or invalid data, an offline controller, an unavailable cloud session or a reading aged 20 seconds or more. Use capture time when present and the receipt time of that measurement otherwise. This view is read-only.

Build and automated checks passed. The owner reported a successful 4.1.0 upload on 06 November after resolving local Config.h declarations and entering DFU mode. Physical checks of all 12 feedback inputs and loss of feedback remain under ACC-FCH.

08 / EMBEDDED DESIGN

Opta firmware architecture

FW-01: Development evolved the supplied v3.3 sketch into modular firmware. Current source version is 4.1.0-feed-chains, telemetry schema 2, built with Arduino IDE or Arduino CLI.

Module	Responsibility
Config.h	Scoped identity, topic/timing constants, private credential placeholders and temperature mapping.
FirmwareTypes.h	Shared frame and sensor-reading structures.
IoManager	I1 conversion, onboard digital inputs, AFX00005 feedback, relay/LED states and connection indications.
TimeSync	Non-blocking NTP, retries, response validation and UTC capture.
TelemetryBuffer	Bounded 120-frame oldest-first RAM queue and dropped-frame counter.
TelemetryCodec	Versioned JSON encoding; null timestamps/temperature and replay diagnostics.
Main .ino	Wi-Fi/MQTT reconnect loop, five-second capture, command parser and queue drain.

Runtime requirements

- FW-02: Capture every 5 seconds even during network loss. Sequence numbers identify capture order, not publish order.

- FW-03: Reconnect Wi-Fi/MQTT without replacing the captured values of queued frames. After reconnect, drain oldest first.

- FW-04: Initialize relay outputs off on boot. This startup behavior is not a full equipment fail-safe design.

- FW-05: Keep field credentials out of Git. Populate only a protected commissioning copy.

Build baseline and known field evidence

Verified toolchain: Arduino CLI 1.4.1, arduino:mbed_opta:opta core 4.6.0, ArduinoMqttClient 0.1.8 and Arduino_Opta_Blueprint 0.2.9. Identity mismatch was corrected by the owner. NTP success was confirmed after the Opta/Mbed UDP endPacket byte-count handling fix. Final commissioned binary checksum and full field test record remain required.

09 / MESSAGE SPECIFICATION

Telemetry and command data contract

MSG-01: Preserve the original outputs, leds, i1 and inputs objects. Firmware metadata is additive. The following synthetic frame illustrates the current contract; captured_at is Unix UTC seconds, not an ISO string.

Illustrative frame; not a live measurement.


```
{
"schema": 2, "firmware": "4.1.0-feed-chains",
"farm_id": "elmajd-poultry", "house_id": "house-01",
"device_id": "opta-control-01", "boot_id": "example-boot-01",
"seq": 12, "captured_at": 1793449800,
"up_s": 345, "rssi": -45, "replay": 0,
"buffer": {"pending": 0, "dropped": 0},
"outputs": {"D0":0,"D1":1,"D2":0,"D3":0},
"leds": {"L0":1,"L1":0,"L2":0,"L3":0},
"i1": {"volts":5.00,"temp_c":25.0,"fault":0},
"inputs": {"I2":0,"I3":1,"I4":0,"I5":0,"I6":0,"I7":1,"I8":0},
"feed_chains": 2081
}
```


Field / rule	Meaning
schema / firmware / identity	Version and registered farm/house/device attribution.
boot_id + seq	Deduplication across reconnects and sequence resets at reboot.
captured_at / up_s / rssi	Source UTC, source uptime and Wi-Fi signal quality.
replay / buffer	Historical replay flag, pending frames and overflow loss counter.
feed_chains	Optional expansion mask. 2081 reports I1, I6 and I12 high.
Faults / unknowns	Before time synchronization captured_at is null. Faulted temperature is null with fault=1; do not substitute a valid zero temperature.

Plain-string command grammar

MSG-02: D0-D3, LED0-LED3 or ALL, followed by ON, OFF, TOGGLE, 1 or 0. The HTTP wrapper is JSON, but the actual MQTT payload is plain text. QoS is 0 and retain is false. No command ID, expiry or device acknowledgement is supplied by this grammar.

Broker acceptance is not evidence of execution. QoS 0 is best effort; command retry/TOGGLE must not be treated as idempotent equipment operation.

10 / DATA QUALITY

Time, freshness and resilience

Requirement	Implemented behavior
TIME-01: Source time	captured_at comes from NTP UTC when valid; received_at is cloud receipt; recorded_at prefers source capture and otherwise uses receipt.
TIME-02: Production calendar	All production/flock/carton dates and daily/weekly grouping use fixed GMT+1 (Etc/GMT-1), without DST.
TIME-03: Identity / latest	Match enabled device/client/topic registration. Older captured frames may enter history but cannot replace a newer device_latest state.
TIME-04: Uniqueness	Current-frame key is farm_id, device_id, boot_id, seq; legacy frames retain compatibility deduplication.
TIME-05: Alarm freshness	Unique, matched, non-replay frames within 30 seconds of receipt with advancing source time qualify. Receipt-time fallback remains when source time is absent.
TIME-06: Persistence	A fresh-sample gap over 30 seconds resets pending duration/consecutive-sample evidence. Active events require a valid recovery sequence.

Device buffer limitations

BUF-01: Capacity is 120 frames, approximately 10 minutes at five-second sampling. Overflow drops the oldest frame and increments buffer.dropped. The queue is RAM-backed and is lost on power cycle. No persistent flash queue is claimed.

BUF-02: QoS 0 provides no broker-level delivery guarantee. Cloud deduplication prevents duplicate inserts; it does not recover an unobserved frame or prove that every attempted publish reached the database.

Offline rule

OFF-01: The server evaluates no fresh accepted measurement for 5 minutes as offline, independently of MQTT status. A newly registered device receives a five-minute grace period. Minute scheduling can add up to about one minute to a timer deadline, excluding network/provider delays.

> Online status messages or replayed backlog do not prove a current fresh measurement. The Android UI must retain last values with stale/source-time context rather than presenting old values as current.

11 / BACKEND SPECIFICATION

Cloud storage and application APIs

Data group	Records / responsibility
Organization / devices	farms, poultry_houses, farm_members, devices; farm membership, logical identity and registered topics.
Telemetry	telemetry raw unique frames; device_latest source-time-guarded current state; telemetry_five_minute chart buckets.
Production / flock	daily_production; flock_settings current count; immutable flock_adjustments and business audit.
Packaging	carton_receipts; calculated carton_balance/opening reference/shortfall.
Alarms / notifications	alarm_settings, alarm_events, alarm_audit, alarm_configuration_audit, alarm_deliveries; phone registrations and provider attempts.
Protected internals	Alarm clocks/state, recovery throttles, worker configuration and encrypted Vault secrets.

Cloud requirements

- DB-01: ingest-mqtt validates the protected webhook, client/topic and telemetry attribution before service-only writes. Invalid or disabled identities must not mutate telemetry.

- DB-02. Keep the latest sample per device and UTC five-minute interval, including temperature, voltage and onboard binary states. Update that row directly from the incoming sample. Store older replay in raw history and preserve the newer chart reading. Keep the sample count.

- DB-03: Client reads require user authentication and farm-scoped RLS. Privileged write RPCs must verify identity, membership and role explicitly.

- DB-04: Business changes run transactionally; invalid dependent history rolls back the full change. Audit records are not directly editable by client applications.

- DB-05. Realtime updates shared state, including the feed chain reading in device_latest. Authenticated REST reconciles missed events and reconnects. DB-06. Retain feed chain readings in telemetry.raw_payload and device_latest.raw_payload with the receipt timestamp of the individual frame.

Shared storage is implemented. Exact retention, backup service level, restore drill, archival/export and disaster-recovery objectives remain open. Cloud storage must not be described as an unconditional permanent-retention guarantee.

12 / APPLICATION SPECIFICATION

Android platform and screens

APP-01: ElMAJD Automation 1.13.1 (29), package com.mahdi.poultrydashboard. Minimum Android API 26; compile/target API 35; Java 17, OkHttp and Firebase Messaging. The UI is bundled HTML/CSS/JavaScript in a native WebView, backed by native authenticated cloud clients.

Tab	Required / implemented content
Overview	Live temperature, latest production/rate with date, current flock, daily/weekly production performance and interactive available-carton tray.
Status	Temperature, separate cisterns I2-I4, silos I5-I6, interactive cooling I7, eight relay/LED fan-control indicators and 12 feed chain states grouped by row and level. Tap supported equipment for history.
3D en direct	Interactive house model with eight fans, 12 chains and two cooling pads; live states, equipment selection, camera views and accessible status list. See APP-3D on pages 14-16.
Alarms & History	Active/recovered/acknowledged events, evidence times, acknowledgement and notification-tap navigation.
Production	Date, Stock, Sortie and mortality; server preview, history pages, audited editing and role-limited deletion. No direct production or actual-flock input.
Configuration	Dated flock changes, categorized reason/reference, carton receipts/adjustments, temperature thresholds, account password and language.
Connection	Cloud session and Realtime diagnostics, source/receipt freshness, controller state, sequence/uptime/RSSI, push/Gmail readiness and self-only tests; pilot command controls.

Charts and mobile behavior

APP-02: Sensor history is the latest 24 hours in five-minute cloud readings selected under DB-02; scale adapts to observed min/max. Missing/invalid data remains a gap. Binary states remain 0/1 rather than inferred volumes. Production charts switch daily/weekly; weekly mortality is summed and production rate is the average of recorded daily rates.

APP-03: French is default with a persistent English option. Provide 44-pixel targets, visible focus, reduced motion, status announcements and accessible measurement/range summaries. Physical TalkBack, large-text and device acceptance remain pending.

Earlier 12-hour UI text belongs to historical versions; the current approved chart baseline is 24 hours.

13 / ANDROID FUNCTIONAL REQUIREMENTS

Live 3D visualisation

The dedicated 3D en direct tab presents the poultry house in Android 1.13.1. The model follows the supplied exterior and cage pictures, with four rows and three levels. Dimensions and exact mounting heights remain indicative.

ID	Required and implemented behaviour
APP-3D-01	Show eight fan indicators, 12 feed chains and two side cooling pads. Equipment selection opens its name, live state and source signal.
APP-3D-02	Fans V1-V4 follow outputs D0-D3; V5-V8 follow LEDs L0-L3. Animation represents control state. Independent rotation confirmation remains deferred.
APP-3D-03	Chains use the existing 12-bit feed_chains reading. Expansion I1-I3 serve row 1, I4-I6 row 2, I7-I9 row 3 and I10-I12 row 4. High confirms running.
APP-3D-04	Both pads use the base Opta inputs.I7 auxiliary contact, as confirmed by the owner. High makes both pads blue with water animation. Separate pad faults cannot be identified.
APP-3D-05	Green indicates a commanded-on fan or running chain. Blue indicates active cooling. Red indicates a valid zero state. Grey indicates unavailable feedback. Always provide text labels.
APP-3D-06	Expire readings after 20 seconds, even without another event. Prefer captured_at; otherwise use received_at for base signals and feed_chains_received_at for chains. Invalid masks or missing signals remain unavailable. Cloud or device-offline events clear live states.
APP-3D-07	Support touch rotation, pinch zoom, two-finger pan, zoom/reset buttons and direct equipment selection. Provide overview, top, fan-wall and pad views; retain keyboard and accessible list interaction.
APP-3D-08	Bundle the renderer and model inside the APK. Reuse authenticated native telemetry with no added cloud query or subscription. Load on first visit, pause outside the tab or hidden app, cap pixel ratio at 1.5 and animation at 24 fps, respect reduced motion and retain status buttons if WebGL 2 fails.

Interaction and permissions

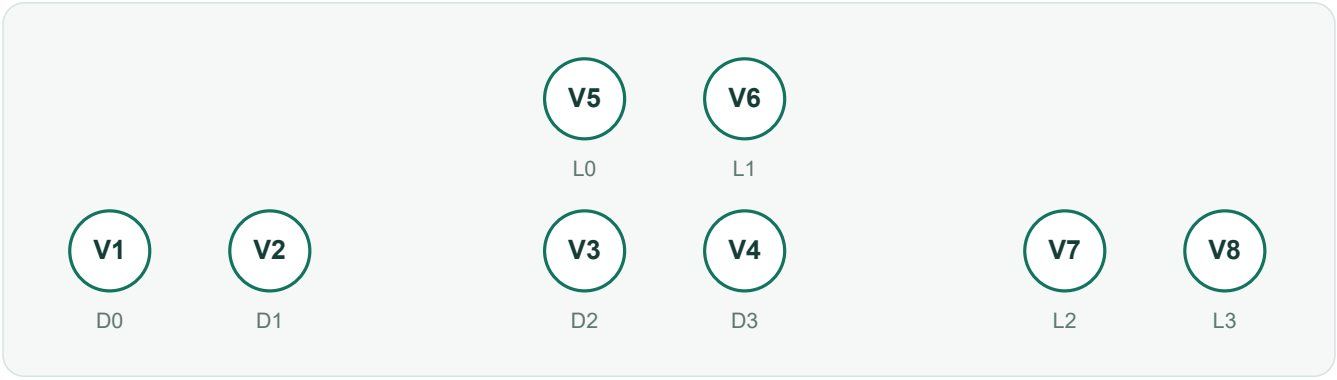
Model selection displays equipment details. Pilot commands continue through the existing authenticated and supervised Connection workflow. Farm membership and existing permissions govern the data delivered to this view.

ACC-3D. On a physical phone, compare all 22 indicators against field signals; test I7 high/low, telemetry loss and recovery, camera selection and rotation, language, reduced motion, background/return, and WebGL fallback. Record the APK version and observed results.

14 / 3D EQUIPMENT SCHEDULE

Fan arrangement and signal mapping

FAN-3D-01. Arrange two horizontal fans on the left, a central 2 x 2 block, and two horizontal fans on the right. The implemented side pairs align with the lower central row. Display numbers preserve the existing signal mapping.



Two horizontal fans + central 2 x 2 block + two horizontal fans

Indicator	Position on the wall	Telemetry source
V1	Left pair, left	outputs.D0
V2	Left pair, right	outputs.D1
V3	Central block, lower left	outputs.D2
V4	Central block, lower right	outputs.D3
V5	Central block, upper left	leds.L0
V6	Central block, upper right	leds.L1
V7	Right pair, left	leds.L2
V8	Right pair, right	leds.L3

Fan camera view

FAN-3D-02. Face the end wall directly and hide other building objects so cages cannot obscure the fans. Fit all eight fans and labels to the viewport and refit on orientation change. Place zoom controls below the wall. Selecting a fan card opens this view and highlights the fan; other views restore the building.

ACC-3D-FAN. Automated checks select all eight fans at 320 px and in CI. On the installed phone, verify the 2 + (2 x 2) + 2 layout, readable V1-V8 labels, visible right-hand pair, camera reset and physical signal-to-fan assignment.

15 / ANDROID INTERFACE EVIDENCE

3D reference views and verification

These screenshots show the developed Android WebView interface from version 1.13.1 using illustrative preview readings. The installed app receives live values through the existing Supabase-to-native telemetry path.

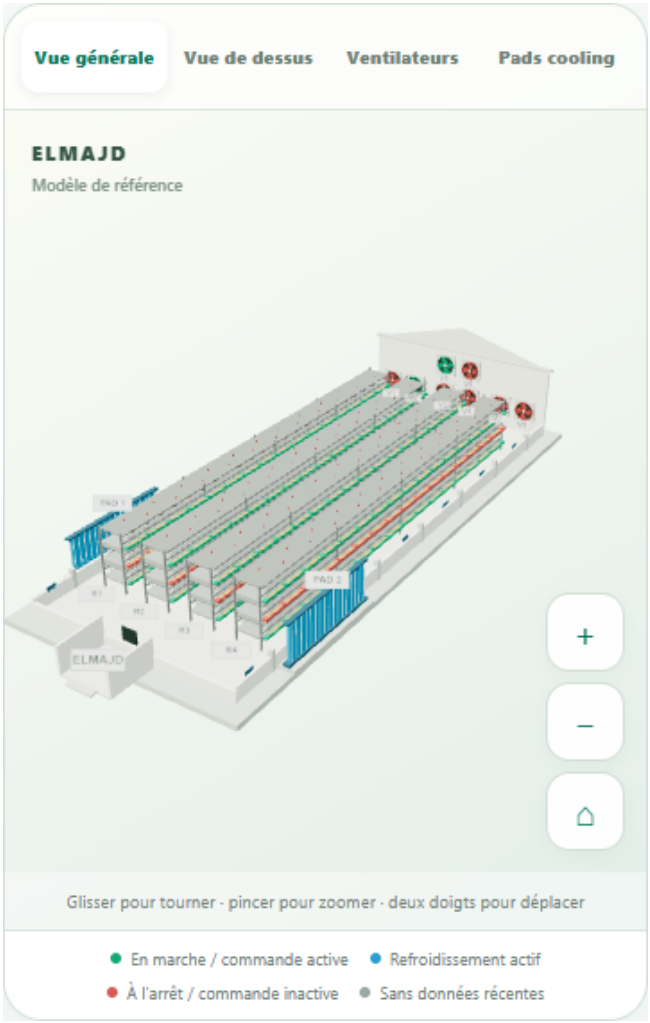


Figure 1. Open-roof house, four rows with three levels, eight fans and two active blue pads. Preview readings are simulated.

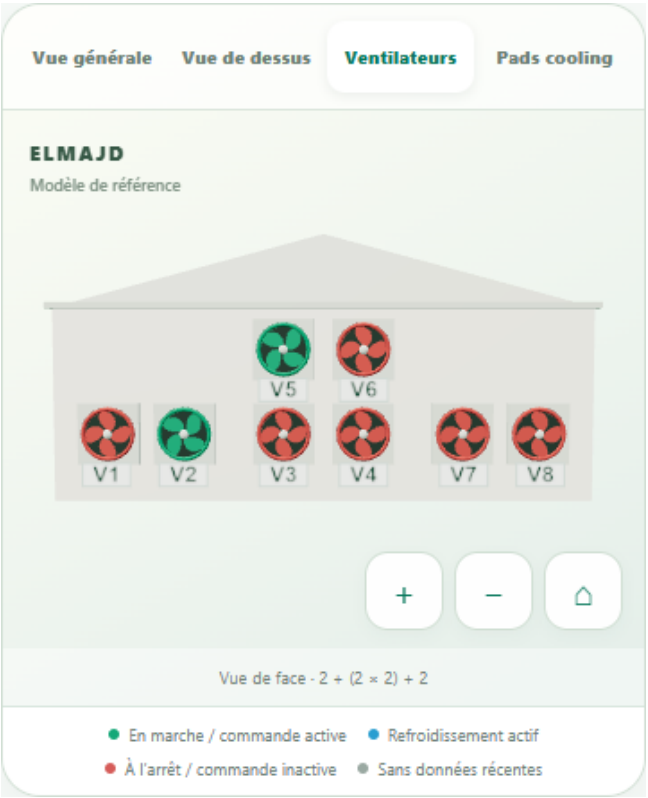


Figure 2. Front fan view showing the corrected arrangement, individual labels and camera controls below the wall. Preview readings are simulated.

Recorded software checks

tests/live3d-ui.cjs checks the exact I/O mapping, all eight fan selections, chain zero/invalid/stale values, shared pads, automatic expiry, cloud/device loss, cameras, French/English, tab lifecycle, file loading and WebGL fallback. tests/phase5-ui.cjs checks seven-tab navigation and existing account, production, carton, role and chart workflows.

Local checks passed at 320 px and 390 px. GitHub Actions 42356825403 passed UI tests, debug compilation, release compilation and pilot signature verification. Packaged JavaScript and CSS were hash-compared with tested sources.

Physical Android installation, actual mounting heights and all 22 field-signal matches remain acceptance checks under ACC-3D. Retain the distinction between fan control state, chain running feedback and the pads' shared auxiliary feedback.

16 / AUTHORIZATION REQUIREMENTS

Accounts, sessions and permissions

AUTH-01: Sign in with Supabase email/password. Android no longer requires direct MQTT credentials in the cloud mode. Broker client ID, username/password and topics are device/server deployment settings; cloud and broker accounts are separate.

Action	Owner/admin	Manager
Read authorized farm data / acknowledge alarms	Yes	Yes
Create or edit production, including historical entries	Yes	Yes
Delete production	Yes, audited	No
Initialize / adjust flock; configure thresholds	Yes	Yes
Add or edit dated carton receipts	Yes	Yes
Delete carton receipts	Yes, audited	No
Send supervised pilot commands	Yes	Yes
Read/update other farms without membership	No	No

Password management and remembered login

AUTH-02: Remembered credentials are encrypted with Android Keystore; backup is disabled to avoid copying installation identity/credentials. Signed-in password changes verify the current password first and update remembered login if enabled.

AUTH-03: Recovery uses a Gmail-delivered Supabase single-use code entered inside the app, not a localhost reset link. Confirmed, eligible owner/manager accounts receive mail; responses are generic, with one request/minute and three/hour per normalized address. Codes and new passwords must never enter logs.

AUTH-04. The command endpoint checks the authenticated user's farm membership and permits the owner and manager roles. Android uses the cloud session to enable sending during a Realtime reconnect. Pilot commands remain supervised.

Multi-farm membership is supported by the data model and read authorization. Current command routing is configured for the single pilot farm/topic; full multi-farm selection/routing remains future work.

17 / BUSINESS SPECIFICATION

Production and mortality rules

Requirement	Authoritative behavior
PROD-01: Inputs	Production date and nonnegative whole Stock, Sortie and mortality. Inputs must fit the supported integer range (0-2147483647).
PROD-02: Calculated eggs	Stock(today) + Sortie(today) - Stock(latest earlier saved date). With no earlier record, calculated eggs = 0.
PROD-03: Missing days	Use the latest available earlier record, not an assumed zero stock and not a requirement for an immediately preceding calendar date.
PROD-04: Flock/rate	Deduct mortality from the day's available flock. Rate = calculated eggs / day's remaining flock x 100, rounded to 2 decimals. Zero flock permits only zero eggs and rate 0.
PROD-05: Validation	Reject negative calculated production, production above remaining flock, excessive mortality, future dates and duplicate farm/date.
PROD-06: Historical change	Recalculate affected stock predecessors, mortality/flock snapshots, rates and cartons atomically. Reject any change that makes a later record invalid.

Worked example

Previous stock=1,000, current Stock=1,250, Sortie=500: production=750 eggs. If pre-mortality flock=1,002 and mortality=2, remaining flock=1,000 and daily rate=75.00%. This is an illustrative calculation, not live farm data.

Preview and history

PROD-07: The read-only preview reads full saved history, including dates outside the loaded phone page. Saving revalidates the transaction; a preview is not a reservation. Stale responses cannot re-enable an invalid save.

PROD-08: History loads in 30-record cursor pages without replacing Overview's latest records. Owner and manager may edit historical Stock/Sortie/mortality; production date stays fixed on edit. Changing a date requires owner deletion/recreation. No month-end lock is implemented; historical corrections remain audited.

18 / BUSINESS SPECIFICATION

Dated flock configuration

FLOCK-01: Actual flock belongs in Configuration. Enter effective date, actual count after the change, category and note/reference. Mortality belongs in Production and is not a second manual flock adjustment.

Category	Meaning / validation
initial	Initial flock and start date before first production; cannot reinitialize an existing flock.
received	Birds received; must increase current count.
transfer_in	Inbound transfer; must increase current count.
transfer_out	Outbound removal/transfer; must decrease current count.
correction	Reasoned correction of count; note required.
reference	System-imported zero-delta migration baseline; not an additional flock movement.

Transaction and audit rules

- FLOCK-02: Compute movement delta from entered actual-after count minus current count. Preserve old/new counts, delta, effective date, actor, category, note and request ID.

- FLOCK-03: Apply delta to relevant production flock snapshots on/after the effective date; preserve earlier dates and recalculate later rates in one transaction.

- FLOCK-04: Historical production before a later movement must exclude that future movement. Reject production before an explicit initial flock date.

- FLOCK-05: Repeating the same request ID with identical data is idempotent; reusing it with different data fails. Clients cannot edit/delete ledger rows directly.

- FLOCK-06: Owner/admin and manager use the authorized RPC. Direct authenticated writes to flock_settings are revoked.

Upgrade to Android 1.11.0 for flock configuration: older APKs use a superseded direct-table write path. The reference migration preserved existing production rows/totals and the current flock; full business reconciliation remains an owner acceptance test.

19 / BUSINESS SPECIFICATION

Carton inventory and opening reference

PACK-01: One carton holds 30 eggs. Consumption is ceil((calculated daily eggs / 30) x 1.10). The fixed 10% allowance is applied before rounding upward. Example: 750 eggs -> 27.5 -> 28 cartons.

Requirement	Current rule
PACK-02: Opening reference	The earliest dated valid carton receipt establishes the inventory reference. No receipt means available stock is 0 and no consumption is counted yet.
PACK-03: Start counting	Exclude production on or before that opening date. Count consumption only from later production dates.
PACK-04: Daily balance	Process dated receipts and consumption chronologically. Available = max(previous available + incoming - used, 0).
PACK-05: Shortfall	Add max(used - previous available - incoming, 0) to a separate cumulative uncovered-use figure. Never display negative available stock.
PACK-06: Later receipts	Later deliveries add stock; they do not repay previously uncovered consumption. Same-day incoming is available for that day's counted use.
PACK-07: Receipt lifecycle	Configuration accepts positive whole cartons, date and note. Owner/manager add/edit; owner alone deletes. Preserve audit; additive retries are idempotent.

Inventory example

Opening date: receipt 10 cartons; that day's production is excluded. Next day: 30 eggs consume 2 cartons -> balance 8. Later usage of 12 leaves balance 0 and shortfall 4. A subsequent receipt of 5 gives balance 5; historical shortfall stays 4.

Editing/deleting the earliest receipt may move or remove the opening reference and change the entire derived balance. The user must review the recalculated result. Shortage is advisory and does not block a valid production record. No separate automatic packaging push/email alarm is claimed by the current implementation.

20 / ALARM SPECIFICATION

Approved alarm policy

ALM-01: Owner-approved pilot defaults operate 24 hours/day, 7 days/week without quiet hours. Both pilot users receive alerts and may acknowledge. Temperature warning/critical thresholds are configurable per house; the listed timing constants are currently fixed.

Condition	Activation	Recovery	Reminders / email
Temperature warning	Valid T >=30 C for 5 min	T <29 C for 5 min	Push every 30 min if unacknowledged; no email.
Temperature critical	Valid T >=33 C for 2 min	T <32 C for 5 min	Push every 10 min; email after 10 min unacknowledged.
Controller offline	No fresh accepted sample for 5 min	One fresh accepted sample	Push every 15 min; email after 10 min.
Temperature fault	Fault, missing/out-of-range T for 2 consecutive samples	2 valid consecutive samples	Push every 30 min; no email.
Cistern/silo low	I2-I6 = 0 continuously 5 min	Affected input = 1 continuously 5 min	Push every 60 min; no email.
Cooling unavailable	Valid T >= warning and I7=0 for 2 min	I7=1 for 2 min, or valid T < warning minus 1 C	Push every 10 min; email after 10 min.

State and configuration rules

- ALM-02: Maintain one open event per farm/device/condition; retain evidence, activation, threshold snapshot, acknowledgement and recovery.

- ALM-03: Acknowledgement suppresses routine repeats and records user/time; it does not clear an active physical condition. Recovery follows the valid recovery rule.

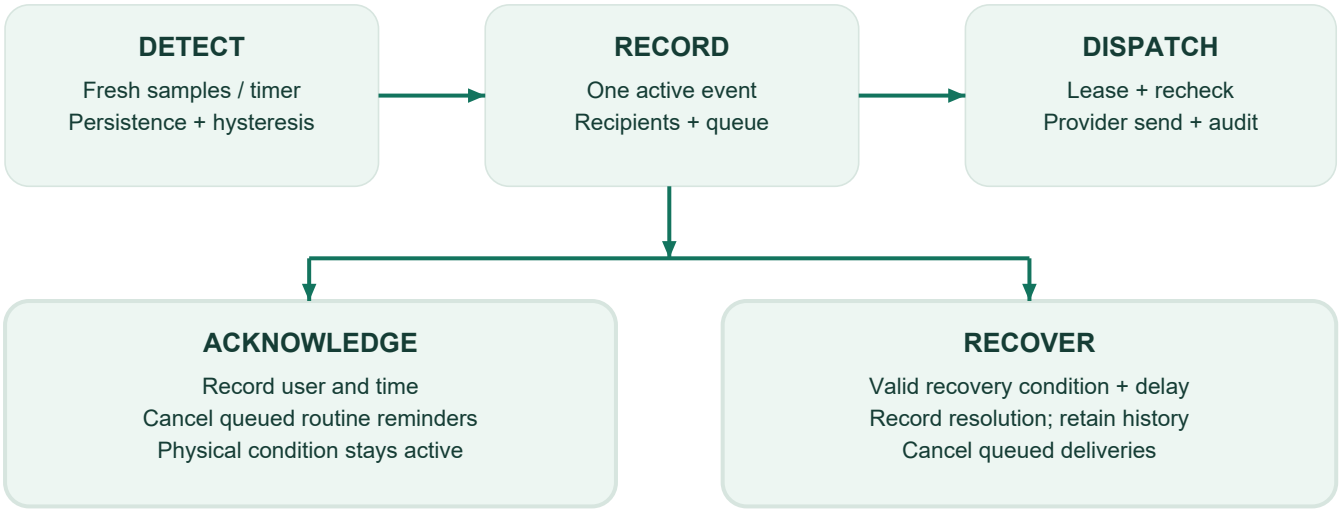
- ALM-04: Threshold changes require owner/manager authorization and before/after audit; reset pending persistence evidence. Active events retain their original threshold snapshot.

- ALM-05: Do not raise fan-failure alarms without independent run feedback. I7=0 alone is not a cooling failure. Low-temperature and SMS policies remain deferred.

The state machine evaluates fresh samples; minute maintenance handles offline, reminders and escalation. Push scheduling is not a guaranteed immediate phone receipt.

21 / SERVER WORKFLOW

Notification delivery and lifecycle



NOT-01: The active dispatch-alarms worker processes push and email independently. Minute cron invokes maintenance and dispatch. Durable queue leases, recipient/state rechecks and provider attempt records support retry and recovery.

Firebase Cloud Messaging

NOT-02: Register each phone against its authenticated user/farm and installation. Synchronize token rotation and permission state; disable invalid tokens. Recheck authorization before sending, record per-phone results and avoid repeating a successful phone while another retries. Native messages open Alarms; local logout suppresses messages immediately.

Gmail fallback and recovery mail

NOT-03: Gmail API uses the authorized sender with send-only scope; protected offline credentials reside in Vault. Active, unacknowledged eligible alarms queue fallback after 10 minutes independently of push acceptance. Temporary failures use bounded retries; record provider IDs. Account recovery uses the same mail adapter.

NOT-04: OAuth now shows Production, approved sender/privacy/terms pages are live, and fresh post-publication consent replaced the Test-mode grant. Formal Google verification is not complete. Refresh grants can still be revoked or require reauthorization.

Owner-confirmed evidence: FCM self-test received; controlled scheduled Gmail fallback received in spam. API acceptance is not human receipt. Already in-flight provider messages cannot be recalled, and a lost response can cause a duplicate retry. App-closed/two-phone/OEM acceptance and normal inbox placement remain pending.

22 / SECURITY BASELINE

Security and release requirements

Requirement	Implementation / remaining condition
SEC-01: Transport / secrets	TLS/HTTPS/WSS; broker/API/webhook and provider credentials remain device/server-side. No service-role or provider private key in APK. Public project IDs/publishable keys are not administrative secrets.
SEC-02: Tenant boundaries	RLS for exposed data; member reads and role-checked RPCs. Privileged functions require explicit identity/tenant checks, restricted grants and controlled search paths.
SEC-03: Audit / validation	Production, receipts, flock, thresholds, alarms, command requests and provider attempts are traceable. Clients cannot overwrite audit or assert sent results.
SEC-04: Native client	Keystore-encrypted remembered login, disabled backup, bundled-only WebView, blocked arbitrary navigation/network/file loads and bridge cleanup.
SEC-05: Sender setup	Temporary credential transfer endpoint disabled, rotation/import RPC removed and generated local refresh-credential copy deleted after successful Vault update.
SEC-06: Pilot signer	Named test APK keeps stable pilot/debug signer for update compatibility. That public development signer is not a protected production identity.
SEC-07: Production release	Release signing requires protected ELMAJD_RELEASE_* environment values; normal release fails without them. CI unsigned release is compilation evidence, not an installable signed release.

Residual security work

Rotate potentially exposed development credentials, verify device-specific broker ACLs and private-key backup and ownership, and review advisor exceptions and leaked-password protection.

A different production signing key may prevent an in-place upgrade from the pilot installation. The release migration, account/session behavior and private signing-key handover require owner acceptance before broad distribution.

23 / NONFUNCTIONAL REQUIREMENTS

Reliability and operations

Requirement	Baseline / operational limit
OPS-01: Live data	Five-second device samples when connected; Realtime for interactive refresh, plus authenticated reconciliation. No hard end-to-end latency SLA is approved.
OPS-02: Background alarms	Server storage/evaluation continues with the app closed. Phone notification delivery still depends on network, permission, Google Play services and OEM policy.
OPS-03: Android force stop	An explicit system Force stop can prevent messages until app reopening. It differs from closing/swiping the UI.
OPS-04: Network interruption	Opta buffers up to 120 RAM frames; overflow/power loss can lose telemetry. Physical outage/replay testing is still required.
OPS-05: Cloud durability	Cloud-shared records are implemented. Retention duration, backup frequency, recovery-point/time objectives and an actual restore drill remain unapproved/unverified.
OPS-06: Observability	Use source/receipt freshness, ingest errors, alarm/provider attempts and cron/worker responses. Define operational monitoring, incident ownership and escalation before commissioning.
OPS-07: Localization	French/English and fixed GMT+1 are implemented; retain original user notes and protocol strings. Full field translation/accessibility acceptance remains pending.
OPS-08: Scale	Pilot is one house/controller and two users. Additional devices/farms require routing, ACL, capacity and membership acceptance; do not copy a shared device identity.

Normal support workflow

- Check Connection for cloud session, Realtime and provider readiness; distinguish cloud offline from controller stale.

- Inspect device_latest plus telemetry source/receipt times for sensor issues; inspect alarm_events and delivery attempts for notification issues.

- Use audited application RPCs for corrections; never repair ordinary business entries by untracked direct table edits.

- Keep a credential/key inventory without secret values and record rotation/restore results in restricted operational records.

24 / VERIFICATION SNAPSHOT

Implementation and evidence matrix

Evidence dates and versions refer to the recorded development and delivery checks. Field acceptance is recorded separately.

Requirement group	Evidence	Status
HW / FW / MSG	Modular firmware and pinned toolchain. Owner confirmed identity, NTP and the 4.1.0 upload.	Implemented; remaining field tests.
FCH-01 to FCH-05	Opta 4.1.0 compilation, Android state and mapping checks, webhook validation and database replay fixture passed. Ingestion v10 deployed.	Physical mapping and loss-of-feedback tests pending.
TIME / DB / OFF	Replay guards and latest-value five-minute storage. Interval, duplicate, fault and receipt-time fallback tests passed.	Deployed and verified. Physical outage test pending.
PROD / FLOCK / PACK	phase4_production_packaging.sql, packaging_opening_reference.sql, production_preview.sql, flock_adjustments.sql.	Six total live SQL suites, including alarm/FCM, passed in rolled-back fixtures.
ALM / NOT	phase3_alarm_engine.sql, fcm_delivery.sql; push-worker.mjs, gmail-worker.mjs, mail-flows.mjs.	Provider/unit/transaction tests passed; owner confirmed self-test receipts.
APP / AUTH	Seven-tab and live 3D UI tests passed. All eight fans selectable; exact signal mapping, expiry, local loading and fallback verified. Pilot 1.11.2 sign-in and D0 ON/OFF previously verified.	Android 1.13.1 phone and ACC-3D field acceptance pending.
Release compilation	GitHub Actions 42356825403. Debug build, unsigned release compilation and pilot signature check passed.	Production signing handover pending.
Work tracking	45 task reports posted and statuses read back; 25 achevé, 20 en cours; 13 other tasks unchanged.	Synchronized 01 November 2026; milestones remain open.

Command role enforcement and observed pilot command results are recorded in the 04 November command recovery report. Feed chain verification used test readings.

Detailed evidence and preserved comment IDs: docs/delivery/phase-3-5-completion.md and docs/delivery/clickup-update-queue.json.

25 / DELIVERY PLANNING

Phases and remaining roadmap

Phase	Delivered position	Remaining gate
0 - Baseline	Policies, decisions, criteria, RACI and risks documented.	Final electrical evidence, named technical accountability and credential-risk treatment.
1 - Firmware	Five-second telemetry, NTP, RAM replay and AFX00005 feed chain reads. Firmware compilation passed.	I/O stimulation, trusted temperature comparison, outage/overflow/reboot and commissioned checksum.
2 - Cloud	Registered identity, protected ingest, raw/latest/five-minute chart history and authoritative fresh clock.	Physical outage/replay, retention/backup objectives and restore evidence.
3 - Alarms	Approved state engine, FCM, Gmail Production grant, audit/ack/recovery and providers implemented.	Controlled critical-condition delivery with app closed; two phones, permission/sign-out/OEM and inbox placement.
4 - Business	Stock/Sortie/mortality, rates, dated flock and carton ledgers with audits/roles implemented.	Owner/manager reconciliation against real records and multi-phone acceptance.
5 - Android	1.13.1 with seven tabs, live 3D, feed chain feedback, French/account flows and production previews.	Install/upgrade, ACC-3D field checks, device UAT and production signing handover.
6 - Secured web	Planned; sender notices are published separately.	Supabase Auth/RLS/Realtime client, feature parity, responsive/accessibility and web acceptance.
7 - Commissioning	Planned; earlier automated evidence informs later testing.	Full integration/security, electrical sign-off, operations/restore/training and go-live acceptance.

Dependency sequence

Complete the remaining field and security checks alongside M3-M5 acceptance. Secured web work follows the cloud/role contract, then Phase 7 validates the complete installation. No date or assignee changes are introduced by this specification revision.

> Parent phases and M0-M5 are still in progress. Completing a feature task does not close the phase gate or imply owner approval for production operation.

26 / RELEASE GATES

Acceptance and test obligations

ACC-01: Record tester, date, phone/Android build, input/condition, expected result, actual result and objective evidence. Use isolated fixtures for synthetic business data or alarm simulation; do not inject invented live farm records.

Gate	Required evidence
M1 / M2 field data	Match all I/O terminals; confirm +/-0.5 C chain; validate UTC/source times, identity, outage replay, duplicate prevention and overflow/power-cycle limits.
ACC-FCH / feed chains	Exercise I1-I12 individually. Verify the matching row and level shows En marche on high and À l'arrêt on low. Check Indisponible after loss of the expansion or stale data. Restore feedback and verify the state recovers. Record firmware, APK, terminal, source time and observed result.
M3 notifications	Critical condition follows configured delay; actual push arrives with app closed/phone locked; acknowledgement cancels queued repeats; valid recovery records history. Verify two phones, sign-out/permission denial and actual Gmail arrival.
M4 calculations	First entry=0; missing dates use latest earlier Stock; mortality/rate/flock movements and historical edits reconcile. Test positive inventory, opening reference, zero floor, shortfall, receipt edits/deletes and role denial.
M5 Android	Upgrade to 1.13.1; verify saved sign-in, production, password flows, French/English, TalkBack/large text, charts and background behaviour. Complete ACC-3D and ACC-3D-FAN on pages 14-16.
Go-live	Electrical approval, private signing/key backup, exposed-credential treatment, backup restore, training and recorded residual-risk/command-scope acceptance.

Minimum business example

Initialize test flock=100. Day 1: Stock=10, Sortie=0, mortality=0 -> production=0. Day 2: Stock=20, Sortie=30, mortality=2 -> production=40, flock=98, rate=40.82%. With opening cartons=10 on day 1, day 2 consumption=2, available=8.

Acceptance decisions

Owner accepts outcomes and any justified deviations; manager tests normal operational workflows. A pending result must remain pending. A waived criterion needs a recorded change and residual risk. Physical commands during tests remain supervised.

The 1.11.0 pilot acceptance sheet remains the reference for its original workflows. Use 1.13.1 with ACC-FCH, ACC-3D and ACC-3D-FAN for this revision.

27 / DECISION REGISTER

Open issues and change control

ID / type	Required next action	Accountability
G-01 / command authorization	Owner/manager enforcement implemented. Retain the role denial evidence with the command acceptance record.	Cloud engineer and owner.
G-02 / hardware evidence	Commission the AFX00005 and complete ACC-FCH. Confirm the cabinet schematic, terminal labels and +/-0.5 C temperature accuracy.	Electrical/firmware roles to be named.
G-03 / acceptance	Complete pending M3-M5 tests with actual phones and business records; normal Gmail inbox placement remains open.	Owner and manager.
G-04 / operations	Choose raw telemetry, five-minute readings, business and audit retention, backups, RPO/RTO, restore tests and monitoring/incident owners.	Owner with cloud/database engineer.
G-05 / release security	Rotate exposed development credentials and hand over a private production signing key and protected backup/upgrade plan.	Owner and technical roles.
G-06 / control scope	Before unsupervised control: define command ID/ack/expiry, independent feedback, interlocks, equipment limits and fail-safe design.	Owner and qualified controls reviewer.
G-07 / scaling and web	Build/accept secured web parity; define farm/house/device selection, scoped topics and per-device command routing.	Phase 6/product/cloud.
G-08 / later enhancements	Independent fan-running feedback, persistent buffer, SMS and low-temperature policy remain deferred; packaging push/email alert policy not implemented.	Owner prioritization.

Resolved decisions - do not reopen as blockers

Two-stage temperature policy and FCM/Gmail choice are approved; sender grant is active in Production. Firmware transfer function is the approved baseline; manufacturer/model paperwork is waived for the pilot. Manager create/edit rights, fixed GMT+1 I7 monitoring sufficiency and the AFX00005 terminal mapping and high-signal meaning are confirmed.

Changes to safety scope, business formulas, permissions, schema or retention require a recorded decision, impact review, versioned implementation and regression evidence. Record the branch/commit and acceptance owner; no automatic merge or deployment follows approval of this book.

28 / APPENDIX

References and delivery identifiers

Artifact	Verified identifier
Specification baseline	v0.5 issued 06 November 2026. v0.6 issued 07 November 2026.
Source branch / review	Opta_Feed_Chain_Expansion. Implementation evidence through 8445db3.
Android test build	1.13.1 (29). Source 8445db3. GitHub Actions 42356825403.
Named APK	ElMajd-Automation-1.13.1.apk, 6,183,553 bytes.
APK SHA-256	998fc9aada78a9fbbf90cad53076ca51efd28cfe2afcc42640d2cf749e3246bc
Firmware	opta_mqtt_control_v4_0_phase1, version 4.1.0-feed-chains, schema 2.
Supabase changes	Ingestion v10. Latest-value migration 20261104195046.
Editable specification	specification-v0.6.json and the companion Markdown in docs/specification.
Public sender pages	elmajd-automation.web.app, /privacy, /terms; information/consent notices only.

Implementation references

See the firmware README for the AFX00005 schedule. The 04 November diagnostic reports record the five-minute history change and command recovery. The phase 3-5 completion report records the earlier business, account and notification checks. Live 3D source, build steps and signal mapping are recorded in docs/phase-5/03-live-3d.md; editable scene geometry is app/src/main/scene/live3d-scene.js.

Platform and delivery references

Supabase: Row Level Security

https://supabase.com/docs/guides/database/postgres/row-level-security

Firebase: receiving Android messages

https://firebase.google.com/docs/cloud-messaging/android/receive-messages

Google: OAuth authorization and token lifetime

https://developers.google.com/identity/protocols/oauth2

GitHub: current delivery and acceptance report

https://github.com/MahdiGU/MQTT-OPTA-Dashboard/blob/French_Account_Gmail/docs/delivery/phase-3-5-completion.md

Arduino AFX00005 specification

https://docs.arduino.cc/hardware/opta-digital-exp

Android 1.13.1 build

https://github.com/MahdiGU/MQTT-OPTA-Dashboard/actions/runs/42356825403

This book documents the developed pilot and its residual gaps. It is not an installation wiring drawing, a safety certification, a Google verification certificate or a production release approval.